Theme Switcher with Animated Progress Bar - Kotlin Console App

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1. Aim

To build a console-based Kotlin application that simulates theme switching and basic animation logic using data classes, functions, and loops.

2. Concepts Used

- Data Class
- Functions
- Loops
- When Expression
- Basic Animation using Thread.sleep()

3. Program Code

```
// Data class to represent a Theme
data class Theme(
    val name: String,
    val primaryColor: String,
    val backgroundColor: String
)

fun applyTheme(theme: Theme) {
    println("\nApplying ${theme.name} Theme...")
    println("Primary Color: ${theme.primaryColor}")
    println("Background Color: ${theme.backgroundColor}")
}

fun animateProgressBar() {
    print("\nLoading: ")
    for (i in 1..30) {
        print("")
        Thread.sleep(100)
    }
    println(" Done!")
}

fun main() {
```

```
    val lightTheme = Theme("Light", "Black", "White")
    val darkTheme = Theme("Dark", "White", "Black")
    val oceanTheme = Theme("Ocean", "Cyan", "Blue")

    println("==== Theme Switcher App ====")
    println("1. Light Theme")
    println("2. Dark Theme")
    println("3. Ocean Theme")
    print("Select a theme (1-3): ")

    val choice = readLine()?.toIntOrNull()

    val selectedTheme = when (choice) {
        1 -> lightTheme
        2 -> darkTheme
        3 -> oceanTheme
        else -> {
            println("Invalid choice! Defaulting to Light Theme.")
            lightTheme
        }
    }

    applyTheme(selectedTheme)
    animateProgressBar()

    println("\nTheme Applied Successfully")
}
```

4. Sample Output

```
==== Theme Switcher App ====
1. Light Theme
2. Dark Theme
3. Ocean Theme
Select a theme (1-3): 2

Applying Dark Theme...
Primary Color: White
Background Color: Black

Loading: ################################ Done!

Theme Applied Successfully
```

5. Conclusion

This program demonstrates the use of data classes, functions, loops, and basic animation logic in Kotlin. The Theme Switcher console application simulates theme application and progress animation in a structured and reusable way.